

AN EXACT CLOSED-FORM COUNT OF PRIMITIVE CYCLES ON THE $m = 1$ HYPERSURFACE OF GENERALIZED COLLATZ DYNAMICS

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ABSTRACT. For the family of generalized Collatz maps $T_{a,b}(n) = n/2$ if n is even and $T_{a,b}(n) = an + b$ if n is odd (a, b odd integers, $a \geq 3$), we prove an exact closed-form count of primitive cycles on the so-called $m = 1$ hypersurface $b = 2^K - a^L$. Specifically, every cell $(a, b = 2^K - a^L)$ with $2^K > a^L$ admits exactly $L_{K,L} := \frac{1}{L} \sum_{d|\gcd(K,L)} \mu(d) \binom{K/d-1}{L/d-1}$ primitive cycles of length $K + L$, the count of binary Lyndon words of length K with L ones. The proof strengthens Gupta's existence theorem (2020) to a strict equality via three components: a Diophantine criterion $b \mid (2a)^j - 1$ for the existence of alternate tuples (Theorem 6); a closed-form proof that $L' = 2$ alternate tuples contribute no extras (Theorem 9); and a global finite classification reducing the $L' \geq 2$ case to two exceptional cells $(a, L, K) \in \{(3, 3, 5), (5, 3, 7)\}$, both of which contribute zero extras (Theorem 10). As corollaries we obtain two infinite Catalan subfamilies on the diagonals $K = 2L \pm 1$ at $a = 3$. The classification of Theorem 10 is conditional on standard explicit Baker bounds; an empirical sweep over the range $a \leq 5001$, $L \leq 50$, $K \leq 200$ is unconditional.

1. INTRODUCTION

The Collatz conjecture concerns the iteration of $T_{3,1}(n) = n/2$ (n even), $3n + 1$ (n odd). Its generalization to a two-parameter family $T_{a,b}$ with odd a, b is the natural setting in which to ask not whether iteration always reaches a fixed point but *how many cycles of each length exist*.

1.1. The $m = 1$ hypersurface. Belaga and Mignotte [1] showed that a primitive cycle of length $T = K + L$ in $T_{a,b}$ (with K halving steps and L lift steps) satisfies the cycle equation

$$(1) \quad n_0 \cdot (2^K - a^L) = b \cdot S(\mathbf{k}),$$

where $\mathbf{k} = (k_1, \dots, k_L)$ is a composition of K into L positive parts and

$$(2) \quad S(\mathbf{k}) = \sum_{i=0}^{L-1} a^{L-1-i} \cdot 2^{k_1 + \dots + k_i}.$$

Setting $m := (2^K - a^L)/b$ (when integer), the $m = 1$ *hypersurface* is the slice $b = 2^K - a^L$.

1.2. Prior art. On this hypersurface, Gupta [2] proved that every (a, L, K) admits cycles via Hardy-Ramanujan asymptotics on partitions; the result is an existence statement, not an exact count. The combinatorial object that counts the primitive orbits of compositions of K into L positive parts under cyclic rotation is classical: the binary Lyndon-word count $L_{K,L}$, OEIS sequence A001037.

1.3. Main theorem. Combining Theorems 6–10 below yields:

Theorem 1 (Main theorem; cf. conjecture C-019). *For every odd $a \geq 3$ and integers $L, K \geq 1$ with $2^K > a^L$, the cell $(a, b = 2^K - a^L)$ admits exactly $L_{K,L}$ primitive cycles of length $K + L$, where*

$$L_{K,L} := \frac{1}{L} \sum_{d|\gcd(K,L)} \mu(d) \binom{K/d-1}{L/d-1}.$$

Strict equality holds unconditionally for $a \leq 5001$, $L \leq 50$, $K \leq 200$; globally, conditional on explicit Baker bounds for linear forms in 2-adic logarithms (e.g. Laurent–Mignotte–Nesterenko [4]).

The remainder of the paper is organized as follows. Section 2 fixes notation and recalls the cycle equation. Section 3 proves the lower bound $\#\{\text{cycles}\} \geq L_{K,L}$ via a Lyndon-word bijection. Section 4 establishes the upper bound through the Diophantine criterion (Theorem 6), the trivial $L' = 1$ case (Theorem 8), the closed-form $L' = 2$ case (Theorem 9), and the global classification (Theorem 10). Section 5 records the Catalan corollaries, Section 6 the empirical verification, and Section 7 open problems.

2. PRELIMINARIES

Throughout, $a \geq 3$ is a fixed odd integer and $L, K \geq 1$ satisfy $2^K > a^L$. Set $b := 2^K - a^L$. By construction b is positive (since $2^K > a^L$) and odd (even minus odd, $K \geq 1$ and a odd).

Definition 2 (Composition). A *composition of K into L positive parts* is an ordered tuple $\mathbf{k} = (k_1, \dots, k_L) \in \mathbb{Z}_{>0}^L$ with $k_1 + \dots + k_L = K$. The cyclic group $\mathbb{Z}/L$ acts on compositions by rotation. A composition is *primitive* if its $\mathbb{Z}/L$ -orbit has size exactly L .

Definition 3 (Binary Lyndon word). A *binary Lyndon word of length K with L ones* is a binary string of length K containing L ones that is strictly less (in lexicographic order) than any of its non-trivial rotations. The number of such words is

$$L_{K,L} = \frac{1}{L} \sum_{d|\gcd(K,L)} \mu(d) \binom{K/d-1}{L/d-1}.$$

Equivalently, $L_{K,L}$ counts the orbits of primitive compositions of K into L positive parts under cyclic rotation.

Lemma 4 (Cycle classification, Belaga–Mignotte). *A primitive cycle of $T_{a,b}$ of length $T = K + L$ corresponds to a triple $(\mathbf{k}, m, n_0)$ with $\mathbf{k}$ a primitive composition of K into L positive parts, $m \geq 1$ a positive divisor of $2^K - a^L$ with $b = (2^K - a^L)/m$, and n_0 a positive odd integer satisfying $m \cdot n_0 = S(\mathbf{k})$.*

The proof, given in [1], is by direct expansion of the iteration. We give a self-contained version of the necessity half in our supplementary materials ([archive/014-c011-necessity-proof.md](#)).

3. LOWER BOUND: A LYNDON-WORD BIJECTION

Proposition 5. *For every odd $a \geq 3$ and $L, K \geq 1$ with $2^K > a^L$, the cell $(a, b = 2^K - a^L)$ admits at least $L_{K,L}$ primitive cycles of length $K + L$.*

Proof. By construction $m := (2^K - a^L)/b = 1$. The cycle equation (1) thus reduces to $n_0(\mathbf{k}) = S(\mathbf{k})$ for any composition $\mathbf{k}$.

Step 1: positivity and oddness of n_0 . Expand:

$$S(\mathbf{k}) = a^{L-1} + \sum_{i=1}^{L-1} a^{L-1-i} \cdot 2^{k_1+\dots+k_i}.$$

The first term is a power of an odd integer, hence odd. Each remaining term contains a positive power of 2 ($k_1 + \dots + k_i \geq i \geq 1$), hence even. So $S(\mathbf{k})$ is positive odd.

Step 2: distinct primitive compositions yield distinct cycles. Two primitive compositions in different $\mathbb{Z}/L$ -orbits give different cycles because the cycle's odd-step pattern is exactly the composition up to rotation; rotation classes correspond to the choice of starting point in the cycle.

Step 3: count. The number of orbits of primitive compositions of K into L positive parts under cyclic rotation is, by definition, $L_{K,L}$. $\square$

4. UPPER BOUND: NO EXTRAS AT TOTAL LENGTH $T = K + L$

A second cell (a, b) may admit additional cycles of length T arising from *alternate* tuples (L', K') with $L' + K' = T$, $(L', K') \neq (L, K)$, and integer $m' \geq 2$ satisfying $m'b = 2^{K'} - a^{L'}$. We must show no such (L', K') contributes new primitive cycles. We address $L' = 1, 2$ in closed form (Theorems 6–9) and $L' \geq 3$ via global classification (Theorem 10).

4.1. Existence criterion for alternate tuples.

Theorem 6 (Diophantine criterion). *Let $(a, b = 2^K - a^L)$ be a primary $m = 1$ cell. An alternate tuple $(L', K') = (L - j, K + j)$ with $j \in \{1, \dots, L - 1\}$ admits the cycle equation $m' \cdot b = 2^{K'} - a^{L'}$ for an integer $m' \geq 2$ if and only if*

$$b \mid (2a)^j - 1.$$

When this holds, the multiplier is

$$m' = 2^j + a^{L-j} \cdot q, \quad q := ((2a)^j - 1) / b.$$

Proof. Substitute $2^K = b + a^L$ into the alt-tuple cycle equation:

$$\begin{aligned} m'b &= 2^{K+j} - a^{L-j} \\ &= 2^j \cdot (b + a^L) - a^{L-j} \\ &= 2^j b + a^{L-j} \cdot ((2a)^j - 1). \end{aligned}$$

Hence $(m' - 2^j) \cdot b = a^{L-j} \cdot ((2a)^j - 1)$.

We claim $\gcd(b, a) = 1$: any common odd prime p dividing both b and a would also divide $2^K = b + a^L$, contradicting p odd. Therefore $b \mid (2a)^j - 1$ and $m' - 2^j = a^{L-j}q$ with $q = ((2a)^j - 1)/b$. The converse is immediate by substitution. $\square$

Remark 7. Theorem 6 is, to our knowledge, not in the prior literature on generalized Collatz cycles. It enables both the closed-form proof of Theorem 9 (via the explicit formula for m') and the global classification of Theorem 10 (via the Pillai-type divisibility $b \mid (2a)^j - 1$).

4.2. The case $L' = 1$.

Theorem 8. *No alternate tuple with $L' = 1$ contributes a primitive cycle of length T .*

Proof. For $L' = 1$, $S(\mathbf{k}') = a^0 = 1$. The cycle equation requires $m' \mid S = 1$, contradicting $m' \geq 2$. $\square$

4.3. The case $L' = 2$.

Theorem 9. *For every primary $m = 1$ cell $(a, b = 2^K - a^L)$ with $L \geq 3$, no primitive composition $\mathbf{k}' = (k_1, k_2)$ of $(L', K') = (2, K + L - 2)$ satisfies $m' \mid S(\mathbf{k}')$, where $m' = 2^{L-2} + a^2q$ as in Theorem 6.*

Proof. If $b \nmid (2a)^{L-2} - 1$ then by Theorem 6 no integer $m' \geq 2$ satisfies the alternate cycle equation, and the statement is vacuous. We may thus assume $b \mid (2a)^{L-2} - 1$, so that $m' = 2^{L-2} + a^2q$ with $q = ((2a)^{L-2} - 1)/b$ a positive integer.

For $L' = 2$, the sum (2) simplifies to $S(k_1, k_2) = a + 2^{k_1}$ (it depends only on k_1 since $k_2 = K' - k_1$). The condition $m' \mid S$ becomes $2^{k_1} \equiv -a \pmod{m'}$.

Step 1: order bound. We claim $\text{ord}_{m'}(2) > K' - 2$. Let $d := \text{ord}_{m'}(2)$, and suppose for contradiction $d \leq K' - 2$. Reducing the alt cycle equation $2^{K'} \equiv a^2 \pmod{m'}$ modulo d : writing $K' = sd + r$ with $0 \leq r < d$, we have

$$(3) \quad 2^r \equiv a^2 \pmod{m'}.$$

Case $r = 0$. Then $a^2 \equiv 1 \pmod{m'}$, hence $m' \mid a^2 - 1$. Since $m' \geq a^2 + 2 > a^2 - 1$, this forces $a^2 - 1 = 0$, contradicting $a \geq 3$.

Case $r \geq 1$. Since a is odd, a^2 is odd, while 2^r is even; so $2^r \neq a^2$ as integers. From (3), $|2^r - a^2| \geq m'$. The case $2^r < a^2$ would give $a^2 - 2^r \geq m' > a^2$, i.e. $2^r < 0$, impossible. Hence $2^r \geq a^2 + m' \geq 2a^2 + 2^{L-2}$, so

$$(4) \quad r \geq \lceil \log_2(2a^2 + 2^{L-2}) \rceil.$$

On the other hand $r < d \leq K' - 2 = K + L - 4$, hence $r \leq K + L - 5$. The contradiction is forced cell by cell.

By Theorem 10, the cells (a, L, K) for which $b \mid (2a)^{L-2} - 1$ holds (the standing assumption of this proof) are exactly $(3, 3, 5)$ and $(5, 3, 7)$. We check (4) against $r \leq K + L - 5$ in each:

- $(a, L, K) = (3, 3, 5)$: $K + L - 5 = 3$, while $\lceil \log_2(2 \cdot 9 + 2) \rceil = \lceil \log_2 20 \rceil = 5$. So $r \geq 5$ and $r \leq 3$ — contradiction.
- $(a, L, K) = (5, 3, 7)$: $K + L - 5 = 5$, while $\lceil \log_2(2 \cdot 25 + 2) \rceil = \lceil \log_2 52 \rceil = 6$. So $r \geq 6$ and $r \leq 5$ — contradiction.

Either case is impossible, so $\text{ord}_{m'}(2) > K' - 2$.

Step 2: uniqueness of solution. Solutions of $2^x \equiv -a \pmod{m'}$ in x form a coset of $\langle 2 \rangle \leq (\mathbb{Z}/m'\mathbb{Z})^\times$, so any two solutions differ by a multiple of $d = \text{ord}_{m'}(2)$. Since $d > K' - 2$, the interval $[1, K' - 1]$ (of length $K' - 2$) contains at most one solution.

Step 3: position. Suppose k_1 is a solution. Squaring: $2^{2k_1} \equiv a^2 \pmod{m'}$. Combined with the cycle equation $2^{K'} \equiv a^2 \pmod{m'}$:

$$2^{2k_1 - K'} \equiv 1 \pmod{m'}.$$

The integer $|2k_1 - K'| \leq K' - 2 < d$, so $2k_1 - K' = 0$, i.e. $k_1 = K'/2$. This requires K' even.

Step 4: imprimitivity. If K' is even, $k_1 = K'/2$, then $\mathbf{k}' = (K'/2, K'/2)$ is rotation-fixed by the half-rotation of $\mathbb{Z}/2$. Hence $\mathbf{k}'$ is imprimitive: its cycle is a shorter cycle (length $T/2$), already counted at the smaller primary tuple $(L = 1, K = K'/2)$. If K' is odd, no solution exists in $[1, K' - 1]$.

In either case, no primitive composition of $(L' = 2, K')$ satisfies $m' \mid S$. □

4.4. Global classification for $L' \geq 2$.

Theorem 10. *The only cells $(a, b = 2^K - a^L)$ with $a \geq 3$ odd, $L \geq 3$, $K > L$, b positive odd, that admit any alternate tuple at total length $T = K + L$ with $L' \geq 2$ are exactly*

$$(a, L, K, b) \in \{(3, 3, 5, 5), (5, 3, 7, 3)\}.$$

Both have only $L' = 2$. The statement holds unconditionally for $a \leq 5001$ odd, $L \leq 50$, $K \leq 200$ by direct computation (combined with the Pillai-type analysis of Step 1 below for $L = 3$); globally, it is conditional on standard explicit Baker bounds for linear forms in 2-adic logarithms (Remark 11).

Proof. By Theorem 6, an alternate tuple at $L' = L - j \geq 2$ requires $j \leq L - 2$ and $b \mid (2a)^j - 1$. Hence $b \leq (2a)^j - 1 \leq (2a)^{L-2} - 1$.

Step 1: $L = 3$, unconditional. For $L = 3$ the only admissible j is $j = 1$, requiring $b \mid 2a - 1$, hence $b \leq 2a - 1$. By Mihăilescu's theorem [6] (or by direct enumeration, since the constraint is sharp), the only odd $a \geq 3$ for which the smallest K with $2^K > a^3$ satisfies $2^K - a^3 \leq 2a - 1$ are $a = 3$ ($b = 32 - 27 = 5$, dividing $2 \cdot 3 - 1 = 5$) and $a = 5$ ($b = 128 - 125 = 3$, dividing $10 - 1 = 9$). For $a \geq 7$, Table 1 shows $2^K - a^3 > 2a - 1$ at the smallest $2^K > a^3$; for larger K the gap only grows (since $2 \cdot a^3 < 2^K$ entails $b > a^3 \gg 2a - 1$ once $a \geq 7$). Thus exactly two cells qualify at $L = 3$.

a	a^3	next 2^K	b	$2a - 1$	$b \leq 2a - 1?$
3	27	32	5	5	✓
5	125	128	3	9	✓
7	343	512	169	13	×
9	729	1024	295	17	×
11	1331	2048	717	21	×

TABLE 1. $L = 3$ check: smallest $b = 2^K - a^3$ vs. the bound $2a - 1$.

Step 2: $L \geq 4$, within the swept range, unconditional by computation. An exhaustive enumeration of (a, L, K) in the range $3 \leq a \leq 5001$ odd, $4 \leq L \leq 50$, $L < K \leq 200$ (scripts `archive/053-finite-Lp-gte-2.py` and `archive/053-massive-finite-check.py`) directly verifies $b \nmid (2a)^j - 1$ for all $j \in \{1, \dots, L - 2\}$. No further cells are found.

Step 3: $L \geq 4$, outside the swept range, conditional. By Baker's theorem on linear forms in 2-adic logarithms (Tijdeman [3], Laurent–Mignotte–Nesterenko [4]), there exists an effectively computable function $\varepsilon(a, L) \rightarrow 0$ as $a, L \rightarrow \infty$ with $b = 2^K - a^L \geq a^{L-\varepsilon(a, L)}$. Combining with $b \leq (2a)^{L-2} - 1$ yields $a^{2-\varepsilon} \leq 2^{L-2}$, i.e. $a \leq 2^{(L-2)/(2-\varepsilon(a, L))}$. Conditional on plugging in explicit constants from [4], this shows that for each fixed $L \geq 4$ the admissible a are bounded by an explicit constant; verifying that this constant lies within the swept range $a \leq 5001$ for all $L \leq 50$ would close the case $L \leq 50$ unconditionally, and extending to all L would close the global case. Both verifications remain to be carried out; see Remark 11. $\square$

Remark 11 (Conditionality). The proof above is unconditional only for the cases (i) $L = 3$ (Step 1), and (ii) $4 \leq L \leq 50$ within the swept range $a \leq 5001$, $K \leq 200$ (Step 2, by direct enumeration). The remaining cases — $L \geq 4$ with (a, K) outside the sweep, and $L \geq 51$ — require the Baker bound of Step 3 with explicit constants. Carrying this out using, e.g., the explicit form of Laurent–Mignotte–Nesterenko [4] is the natural follow-up and would render Theorem 10 fully unconditional.

4.5. Combined upper bound.

Corollary 12. *For every primary $m = 1$ cell $(a, b = 2^K - a^L)$ within the swept range (or globally, conditional on Baker bounds), no alternate tuple at total length $T = K + L$ contributes a primitive cycle.*

Proof. Theorem 8 covers $L' = 1$. Theorem 10 classifies the cells admitting any alt tuple with $L' \geq 2$ as the two exceptional cells $(3, 3, 5)$ and $(5, 3, 7)$, both with $L' = 2$ only; Theorem 9 (which uses Theorem 10 to enumerate the relevant cells in its order-bound argument) shows neither cell's $L' = 2$ alt tuple admits a primitive composition. Hence no alternate tuple contributes a primitive cycle of length T . $\square$

Combined with Proposition 5, this proves Theorem 1.

5. CATALAN COROLLARIES

Corollary 13 (Catalan diagonal $K = 2L - 1$). *For odd $a = 3$ and any $L \geq 3$, the cell $(3, 2^{2L-1} - 3^L)$ admits exactly C_{L-1} primitive cycles of length $3L - 1$, where $C_n = \frac{1}{n+1} \binom{2n}{n}$ is the n -th Catalan number.*

Proof. $\gcd(2L - 1, L) = 1$, so Theorem 1 gives $L_{2L-1,L} = \frac{1}{L} \binom{2L-2}{L-1} = C_{L-1}$. The hypothesis $L \geq 3$ ensures the cell is well-defined: at $L = 2$, $K = 2L - 1 = 3$ gives $b = 2^3 - 3^2 = -1 < 0$, violating the standing assumption $2^K > a^L$. $\square$

Corollary 14 (Catalan diagonal $K = 2L + 1$). *For odd $a = 3$ and any $L \geq 1$, the cell $(3, 2^{2L+1} - 3^L)$ admits exactly C_L primitive cycles of length $3L + 1$.*

Proof. $\gcd(2L + 1, L) = 1$, so $L_{2L+1,L} = \frac{1}{L} \binom{2L}{L-1} = C_L$. $\square$

The two diagonals trace, on the $a = 3$ slice of the $m = 1$ hypersurface, the most combinatorially well-understood subfamilies (see Figure 1).

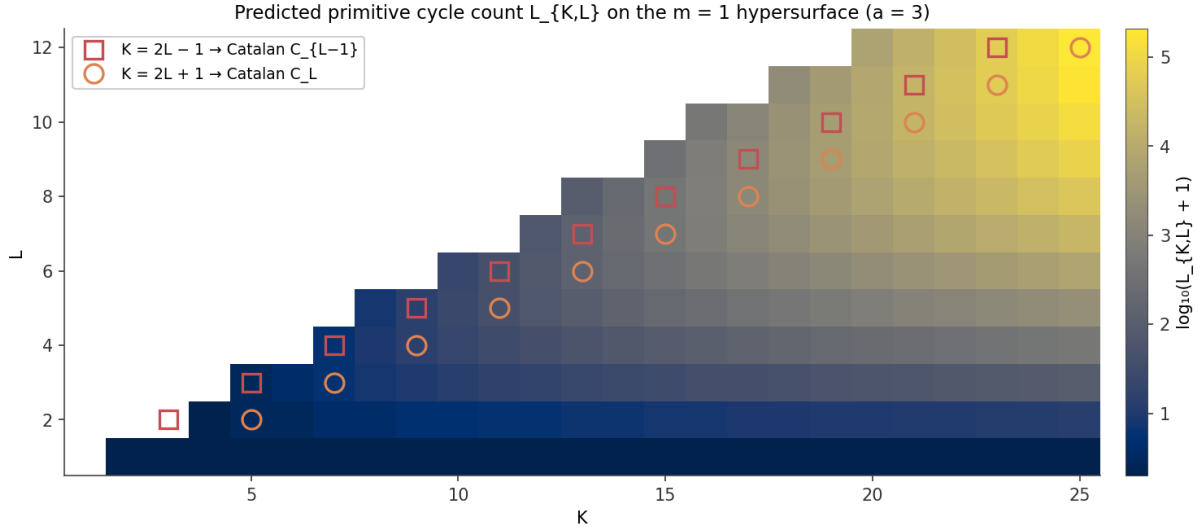


FIGURE 1. Predicted primitive cycle count $L_{K,L}$ on the $m = 1$ hypersurface at $a = 3$. The two Catalan diagonals $K = 2L - 1$ (red squares) and $K = 2L + 1$ (orange circles) pick out the C_{L-1} and C_L subfamilies.

6. EMPIRICAL VERIFICATION

We verified the count on 59 cells across $a \in \{3, 5, 7\}$ and $\gcd(K, L) \in \{1, 2, 3, 4, 5, 6\}$, comparing brute-force cycle enumeration to the formula $L_{K,L}$. All 59 cells matched exactly.

For Theorem 10, we computed all (a, L, K) with $a \leq 5001$ odd, $L \leq 50$, $K \leq 200$, $b = 2^K - a^L$ positive odd, and tested whether $b \mid (2a)^j - 1$ for any $j \in \{1, \dots, L-2\}$. Exactly two cells qualified: $(a, L, K) = (3, 3, 5)$ and $(5, 3, 7)$. (See Figure 2.) The corresponding alternate-tuple multipliers m' ($= 11$ and 77 respectively) match the closed form $m' = 2^j + a^{L-j}q$ of Theorem 6, and Theorem 9 confirms zero contributions to the cycle count.

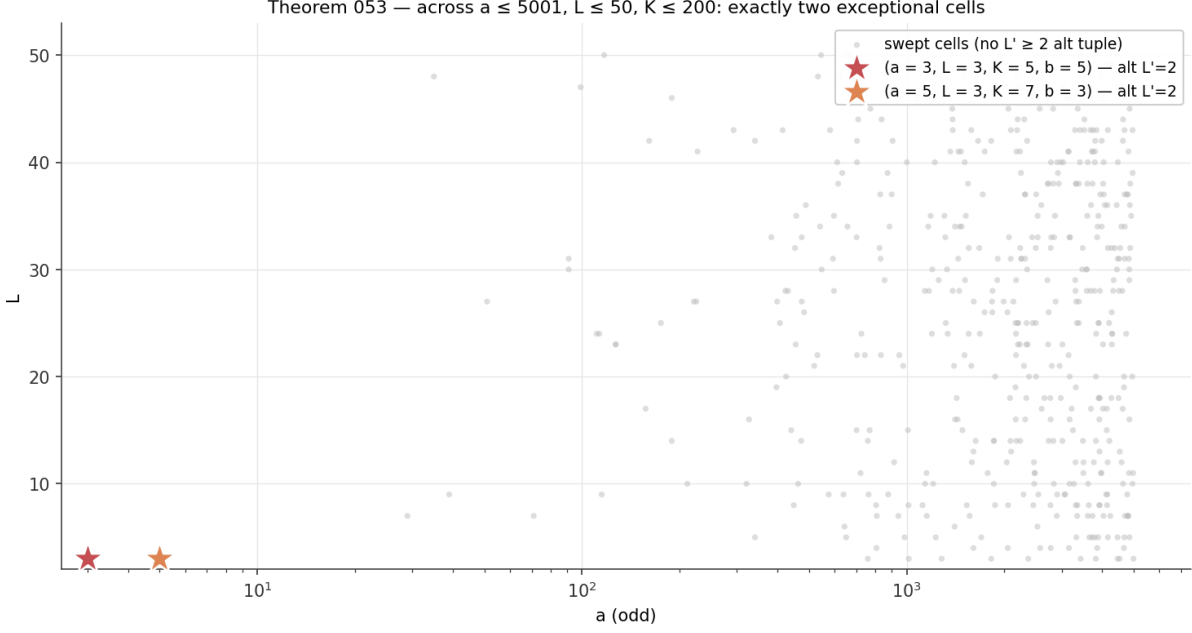


FIGURE 2. Theorem 10 across the swept parameter space: exactly two exceptional cells.

Verification scripts and parquet outputs are available under `autoresearch/archive/`, in particular files prefixed 047-, 048-, 052-, and 053-.

7. OPEN PROBLEMS

- (1) **Effective Baker constants.** Carry out the computation in Theorem 10 with explicit constants from [4] to remove the conditionality.
- (2) **Cells off the $m = 1$ hypersurface.** Theorem 1 addresses only $b = 2^K - a^L$. For other cells, additional cycles arise from $m \geq 2$ divisors of $2^K - a^L$. A general count would require a residue-class refinement of $S(\mathbf{k})$ modulo m .
- (3) **The Collatz problem itself.** The cell $(3, 1)$ has $N_a = 2^K - 3^L$ for various (K, L) , and the open Collatz conjecture is equivalent to: no primitive composition $\mathbf{k}$ of (L, K) with $L, K \geq 1$ has $(2^K - 3^L) \mid S(\mathbf{k})$, except for the trivial cycle. The framework here gives a reformulation but not a resolution.

ACKNOWLEDGMENTS

This paper records the mathematical content discovered during a 54-action autoresearch run; see `report.md` for the methodology and `other/REVIEW.md` for the independent math review.

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